

Projection Folds and Apparent Pair Creation: A Computational Falsification Framework for Hidden-Manifold Models

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(Dated: August 3, 2026)

A smooth trajectory can intersect a constant-time slice twice after its projection into spacetime develops a nondegenerate fold. An observer using spacetime time as the evolution coordinate then sees a particle-antiparticle-like pair appear, even though the underlying trajectory neither begins nor branches. The local statement is exact: the two projected branches have opposite time orientation and preserve a signed intersection number. It is also old in spirit, lying close to Stueckelberg’s worldline interpretation of pair creation. The open question is not whether folds can depict pairs, but whether a local hidden-manifold dynamics can predict their statistics without putting quantum-field-theoretic answers into the projection, initial measure, or selection rule.

We formulate that question as a falsification campaign. Exact monotone, fold, degenerate, and double-fold controls validate branch counting and event location. A conserved Hamiltonian toy model tests how easily generic local dynamics produces coordinate-time folds and serves as a negative control for universal rate claims. A prior-art replication anchors the campaign in Stueckelberg–Horwitz–Piron dynamics. The first physical gate is a held-out Schwinger-scaling challenge: a candidate model must recover the exponent $\exp[-\pi m^2/|qE|]$, conservation, complete outcome accounting, observer covariance, and gauge independence without retuning. We give an implementation plan for MATLAB on a local research server and containerized CPU/GPU ensembles on the National Research Platform. The framework is designed to make a clean negative publishable: a fold may remain a useful kinematic representation even if it fails as a mechanism of quantum pair production.

I. INTRODUCTION

Particle creation is often drawn as a worldline that turns in coordinate time. A single continuous curve, parameterized by an evolution variable independent of spacetime time, can appear on a constant-time slice as two objects with opposite time orientation. Stueckelberg used this construction explicitly in 1941, and modern Stueckelberg–Horwitz–Piron (SHP) dynamics retains an invariant evolution parameter τ independent of the spacetime coordinates [1, 2]. Worldline methods also give a standard semiclassical description of Schwinger production, where closed Euclidean trajectories determine the nonperturbative exponent [3, 4].

The geometric picture invites a stronger hypothesis. Suppose observed spacetime is a projection of a larger state manifold. Could apparent pair production be a stable singularity of that projection rather than a fundamental creation event? The question is mathematically coherent, but it is easy to answer circularly. A model can always be assigned a projection, hidden-state density, or stopping rule that produces the desired number of branches. Reproducing a picture is not the same as deriving a rate.

This paper therefore makes a deliberately limited contribution. It does not propose a completed alternative to quantum field theory. It separates:

1. an exact kinematic statement about projection folds;
2. a dynamical question about the flux of trajectories through a fold set; and
3. a physical question about whether that flux reproduces quantum pair production.

We then specify a computational campaign in which each level has its own witness and falsification bar. The methodology follows a simple rule: a claim may not exceed what the recorded observables can distinguish. In particular, every trajectory is counted. A model is not permitted to discard no-fold, escape, recrossing, or numerically inconvenient outcomes and then interpret the retained subset as a production rate.

II. PROJECTION FOLDS

Let $\mathcal{M}$ be a smooth state manifold and let

$$z : I \rightarrow \mathcal{M} \quad (1)$$

be a smooth trajectory parameterized by an evolution variable τ . A smooth observation map

$$\pi : \mathcal{M} \rightarrow \mathcal{X} \quad (2)$$

produces an observed spacetime trajectory $x^\mu(\tau) = \pi^\mu(z(\tau))$. An observer or detector defines a time functional

$$T_u(x) = u_\mu x^\mu, \quad (3)$$

where u specifies the observer’s time direction. Write $f(\tau) = T_u(x(\tau))$.

Definition 1 (Worldline-time fold). *A point τ_0 is a non-degenerate worldline-time fold when*

$$f'(\tau_0) = 0, \quad f''(\tau_0) \neq 0. \quad (4)$$

The singularity belongs to the composition $T_u \circ \pi \circ z$. Spacetime itself need not be degenerate. Nor does the fold require many hidden dimensions; one independent evolution parameter is enough. Higher dimension earns explanatory weight only if it predicts the distribution or dynamics of folds.

Proposition 1 (Local pair multiplicity). *Let τ_0 be a nondegenerate worldline-time fold with $t_0 = f(\tau_0)$. In suitable local coordinates,*

$$f(\tau) - t_0 = \sigma(\tau - \tau_0)^2, \quad \sigma \in \{+1, -1\}. \quad (5)$$

For $\sigma = +1$, the number of intersections with a nearby constant-time slice changes from zero to two as t crosses t_0 from below. The two regular branches have opposite signs of $f'(\tau)$. For $\sigma = -1$, the time-reversed statement holds.

Proof. The one-dimensional Morse lemma gives the stated normal form. For $\sigma = +1$ and $t > t_0$, the two solutions are

$$\tau_{\pm} = \tau_0 \pm \sqrt{t - t_0}. \quad (6)$$

Their derivatives satisfy $f'(\tau_+) = -f'(\tau_-) \neq 0$. No real solutions exist for $t < t_0$. The case $\sigma = -1$ follows by reversing the direction of the time coordinate. $\square$

Define the signed intersection count on a regular slice by

$$N_{\text{sgn}}(t) = \sum_{\tau_i: f(\tau_i)=t} \text{sgn } f'(\tau_i). \quad (7)$$

A simple fold changes the unsigned count by two and the signed count by zero. If one chooses to interpret orientation as a charge label, opposite labels follow. That interpretation is an additional physical hypothesis; the topological result itself establishes only opposite orientation.

The local separation of branches is also diagnostic. If an observed spatial coordinate has nonzero first derivative at the fold, $x(\tau) = x_0 + v(\tau - \tau_0) + \dots$, then

$$\Delta x(t) = 2|v|\sqrt{|t - t_0|} + o(\sqrt{|t - t_0|}). \quad (8)$$

This square-root law is an exact instrument target.

III. FROM A PICTURE TO A MECHANISM

A fold permits an apparent pair event. It does not determine when events occur, with what probability, or with what energy and quantum numbers. A candidate mechanism must specify at least

$$(\mathcal{M}, \mathcal{D}, \pi, \mu_0), \quad (9)$$

where $\mathcal{D}$ is a local dynamics on $\mathcal{M}$ and μ_0 is the initial-state measure. The critical set is

$$\mathcal{F}_u = \left\{ z : \frac{d}{d\tau} T_u(\pi(z(\tau))) = 0 \right\}, \quad (10)$$

and the putative production rate is a trajectory flux through the nondegenerate part of $\mathcal{F}_u$.

The phrase “anti-hub” is not used here. Near a fold, a projection generally concentrates rather than depletes observed density because the projection Jacobian vanishes. The mathematically relevant objects are projection multiplicity, critical-set flux, and orientation.

A. Anti-circularity conditions

A model is excluded from claim-bearing evaluation if it contains any of the following in explicit or equivalent form:

- a rule that inserts folds at a desired rate;
- the Schwinger exponent in the initial-state weights or retention rule;
- training and evaluation on the same field, mass, and charge region;
- postselection on trajectories that fold or resemble particles;
- a coordinate-time choice transformed without the observer and detector;
- a charge or energy assignment chosen only after inspecting the branches.

A long parameter search can otherwise manufacture almost any radial or exponential law. The scientific object is not the best-fitting hidden geometry, but the held-out predictions of a geometry fixed for independent reasons.

IV. CONTROLS AND CANDIDATE DYNAMICS

A. Exact instrument controls

The measurement stack begins with six controls.

TABLE I. Instrumentation controls. The cubic critical point prevents a zero of $dt/d\tau$ from being treated automatically as a pair fold.

Control	Time map	Required result
N0	$t = \tau$	no folds
P0	$t = \tau^2$	$0 \rightarrow 2$ branches
P1	$t = -\tau^2$	$2 \rightarrow 0$ branches
D0	$t = \tau^3$	degenerate, not a fold
M0	$t = \tau^3 - \tau$	two critical points recovered
S0	circular instanton action	$R_* = m/ qE $, $S_* = \pi m^2/ qE $

Control S0 uses the reduced circular worldline action

$$S(R) = 2\pi mR - \pi|qE|R^2. \quad (11)$$

Its stationary radius and action are

$$R_* = \frac{m}{|qE|}, \quad S_* = \frac{\pi m^2}{|qE|}. \quad (12)$$

It verifies that the rate-analysis code can recover the standard semiclassical scaling before it is applied to a new dynamics.

B. A conserved negative control

We use the toy Hamiltonian

$$H = \frac{1}{2}(p_t^2 + p_x^2 + p_u^2) + \frac{1}{2}\omega_t^2 t^2 + \frac{1}{2}\omega_u^2 u^2 + \frac{\lambda}{4}u^4 + gEtu. \quad (13)$$

The variable u is hidden, τ is the Hamiltonian evolution parameter, and the coordinate-time velocity is $dt/d\tau = p_t$. Folds occur whenever p_t crosses zero with nonzero acceleration.

Equation (13) is intentionally not relativistic and is not proposed as a model of QED. It asks a narrower question: how often does a generic local, energy-conserving system produce projection folds, and how sensitive is the answer to initial-state measure, slicing, and numerical method? A likely result is that folds are easy to obtain but their flux has no universal Schwinger dependence. That negative is useful because it separates the stable geometry of a fold from the far stronger physics of a pair-production rate.

C. Prior-art replication

Before modifying SHP dynamics, the campaign reproduces one published classical pair trajectory and its total conservation accounting [2]. The equations, gauge-field conventions, initial conditions, and numerical tolerances are frozen in a replication record. Failure to reproduce the selected case blocks novelty claims. This stage prevents the proposed framework from quietly rediscovering a known worldline construction under new terminology.

V. THE EXPERIMENT CAMPAIGN

A. PF-0: event-location audit

PF-0 measures branch counts, signed counts, critical-point classification, and Eq. (8). The instrument fails if it misses a simple fold, labels the cubic control as a fold, or changes the signed intersection number outside the sealed numerical tolerance.

B. PF-1: structural stability

Smooth random perturbations are added to the exact fold. The campaign measures the persistence of a nearby nondegenerate critical point, the fitted square-root exponent, and the condition number of event location. Numerical precision and sampling density are swept independently. A positive result is not pair physics; it is confirmation that the code sees the stable singularity it claims to see.

C. PF-2: generic local dynamics

Ensembles of Eq. (13) are run over field-like coupling, oscillator scales, nonlinearity, initial-state distribution, and observer slicing. Every trajectory receives an outcome: no fold, fold, recrossing, escape, or numerical failure. The primary quantities are fold flux, branch lifetime, creation–annihilation balance, and Hamiltonian drift. The result is rejected as a mechanism if an apparent law is explained by the initial measure, discarded trajectories, or coordinate choice.

D. PF-3: Schwinger scaling challenge

The first physical gate uses a candidate dynamics fixed before the sweep. One overall time-volume conversion and, if unavoidable, one rate prefactor may be calibrated on a training region. Contiguous ranges of E , m , and q are held out. The target is

$$\log \Gamma = \text{const} - \frac{\pi m^2}{|qE|} \quad (14)$$

within a preregistered approximation regime.

The candidate is compared with polynomial-threshold, power-law, alternative exponential, and complexity-penalized spline models. It fails if the held-out slope excludes $-\pi$, if a simpler alternative predicts better, if the result is not stable under integration refinement, or if an equivalent of Eq. (14) was embedded in the initial measure or stopping rule.

E. PF-4: conservation and complete accounting

A branch-count change is not sufficient. The underlying model must conserve its stated total quantities. If the background field is dynamical, energy acquired by the projected branches must be supplied by the field. An orientation-based charge rule must be declared in advance and sum to the initial charge. Missing or deleted trajectories are reported as primary outcomes, not ignored as implementation noise.

F. PF-5: covariance and gauge audit

A coordinate-time fold may be observer dependent. The complete setup is therefore transformed: state, fields, projection, and detector time direction. A passive coordinate change must transform event worldpoints without changing invariant rates. Electromagnetic implementations are repeated in gauge-equivalent potentials with the same field tensor. Observer-dependent particle assignments are retained only when a physical detector model predicts the corresponding difference.

G. PF-6: quantum ceiling

Only after the preceding gates pass does the campaign test scalar–spinor differences, multipair statistics, Pauli blocking, interference in pulsed fields, and backreaction. A classical fold model is expected to fail at least some of these tests. Such failure does not refute the fold theorem; it limits the model to a worldline representation rather than a replacement for quantum field theory.

VI. EXECUTION ON ATLAS AND NRP

The reference package contains MATLAB and Python implementations. MATLAB on Atlas is used first because adaptive ODE solvers can locate zero crossings through event functions, and Parallel Computing Toolbox supports parameter sweeps with `parfor` [5, 6]. Atlas also serves as the independent reducer for selected NRP outputs.

The National Research Platform (NRP) is used for large finite ensembles. NRP recommends Kubernetes Jobs for computations that run to completion and prohibits batch jobs that merely sleep while awaiting manual use [7]. Current cluster policy requires resource requests to match measured use and requires equal requests and limits for large populations of pods or jobs [8]. The campaign therefore keeps its full manifest in durable storage and exposes only a bounded live working set. Four bounds are monitored independently:

1. active workers;
2. Kubernetes objects and pending pods;
3. storage and network pressure; and
4. unreduced evidence backlog.

GPU runs begin with one device per job and record device type, CUDA runtime, precision, and image digest. The NRP documentation supports requesting generic or specific GPU resources, but utilization must justify the allocation [9]. The supplied GPU path is a vectorized fixed-step integrator for the toy negative control; it is not used for claim-bearing SHP or Schwinger results until cross-validated against adaptive CPU integration.

Every trial writes an immutable JSON record containing the complete scenario, random seed, projection, observer, numerical method, all outcome counts, conservation residuals, resource use, code commit, container digest, and hashes of inputs and outputs. A reducer may summarize records but may not rewrite them.

VII. WHAT WOULD COUNT AS A RESULT?

Several outcomes would be scientifically useful.

a. Expected negative. Simple folds are structurally stable, generic local systems produce them, but fold flux depends on arbitrary dynamics and initial measure and fails the held-out Schwinger law. The conclusion would be that projection folds are a valid kinematic representation and not a mechanism of QED pair production.

b. Stronger negative. A broad model class can be shown analytically or computationally to produce only power-law or threshold fold flux unless an exponential action or equivalent large-deviation structure is inserted. This could become a no-go result for naive hidden-manifold models.

c. Unexpected positive. One fixed, local, covariant dynamics reproduces Eq. (14), complete conservation, gauge independence, held-out field profiles, and later quantum signatures. That would justify a much stronger theory paper. A fitted $0 \rightarrow 2$ branch plot would not.

VIII. RELATION TO BLACK-HOLE PARTICLE CREATION

The same language may later be applied to Hawking or Unruh settings, where particle content depends on the relation between field modes and an observer’s time coordinate. The relevant benchmark would not be the existence of two branches, but recovery of the exponential null-coordinate mapping, Bogoliubov coefficients, thermal spectrum, and detector response. We leave that extension outside the first campaign. A model that cannot pass the constant-field Schwinger gates has no basis for a black-hole information claim.

IX. DISCUSSION

A fold supplies an economical explanation of why a projected worldline can look like a pair with opposite orientation. It does not explain why electrons have their measured mass, charge, spin, statistics, and production rate. The distinction between depiction and derivation is the center of the proposed campaign.

The likely value of this work is methodological. Hidden-dimensional and emergent models are often judged by whether they can display a desired curve. Here the model must survive exact nulls, positive controls, complete accounting, held-out parameter regimes, independent im-

plementations, conservation, and covariance. A clean rejection leaves the geometry intact while removing an unsupported physical interpretation. A positive result must earn each additional word in the claim.

DATA AND CODE AVAILABILITY

The accompanying source package contains the manuscript, experiment plan, preregistration template, MATLAB controls and ODE pilots, a Python reference implementation with unit tests, and NRP Kubernetes templates. No claim-bearing results are included in this draft.

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